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When soft regulation is not enough: The integrated pollution prevention and control directive of the European Union

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Abstract

Recent debates regarding the effectiveness of regulatory policymaking in the European Union (EU) focus on the merits of soft, non-binding forms of regulation between public and private actors. The emergence of less coercive forms of regulation is analyzed as a response to powerful functional pressures emanating from the complexity of regulatory issues, as well as the need to secure flexibility and adaptability of regulation to distinctive territorial economic, environmental, administrative, and social conditions. In this article we empirically assess the above normative claims regarding the effectiveness of soft regulation vis-à-vis uniformly binding legislation. We draw on an exploratory investigation of the application of the Integrated Pollution Prevention and Control Directive of the EU in four countries. Our study reveals that effectiveness in the application of soft policy instruments is largely contingent upon strong cognitive, material, and political capacities of both state regulators and industrial actors involved in regulatory policymaking. In the absence of those conditions, the application of soft, legally non-binding regulation may lead to adverse effects, such as non-compliance and the "hollowing out" of the systems of environmental permits to industry. In the medium term, such developments can undermine the normative authority of the EU.

Keywords: best available technologies, European Union, industry, pollution prevention, soft regulation.

1. Introduction

Much of the debate on increasing effectiveness and legitimacy of European Union (EU) policies focuses on the merits of soft, non-binding forms of regulation between public and private actors (Mörth 2004; Sabel & Zeitlin 2010; Héritier & Rhodes forthcoming).

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Soft regulation refers to a wide range of quasi-legal instruments that differ from hard law as they lack immediate, uniformly binding, direct effects, precision, and clearly delineated monitoring, dispute settlement, and enforcement authorities (Shaffer & Pollack 2009). These instruments take different forms, including guidelines and recommendations, resolutions, declarations, codes of practice and conduct. The debate about the rise of soft regulation reflects much wider considerations among scholars of international relations, politics, and law regarding the conditions that contribute to the emergence of transnational legalization, its components and types of norms, rules and institutions, and the conditions that influence the behavior of state and non-state actors (Abbott *et al.* 2000; Goldstein *et al.* 2000; Peters *et al.* 2009; Börzel and Risse 2010).

The increasing reliance on soft regulation as a tool of global and regional governance is analyzed as a response to powerful functional pressures emanating from the complexity of regulatory issues, uncertainty over regulatory goals and desirable outcomes, and interdependence between public and private actors at multiple levels of regulatory policymaking (Trubek & Trubek 2005). Rational choice accounts point at the benefits of soft regulation for regulators and regulatees in terms of transaction costs, advancement of political strategies, and resolution of problems of incomplete contracting. Reliance on non-binding, less coercive instruments is analyzed as a deliberate choice of actors because they offer clear benefits over legally binding regulation. In highly heterogeneous regulatory areas non-binding instruments facilitate the need to secure flexibility and adaptability of regulation to distinctive economic, environmental, social, and administrative conditions that cannot be achieved through uniformly binding legislative requirements (Héritier 2003). The lack of uniformly binding obligations allows discretion to local regulators and the industry to define precise regulatory targets especially in policy areas that involve high compliance costs, most notably production processes and biodiversity protection. Therefore, in diverse regulatory regimes, the adoption of less precise, non-binding arrangements contributes to lower transaction costs in negotiations between interested parties and facilitates coordination and cooperation when the option of hard regulation is not available due to high sovereignty or economic costs (Abbott & Snidal 2000). In these terms, soft regulation is often conceptualized as a tool for overcoming initial stalemate in multilateral negotiation and for learning and experimentation to achieve future consensus and as a first step for the development of hard law (Mörth 2004; Sheldon 2000). Social constructivists, finally, point at the contribution of soft regulation to the diffusion of norms among state and non-state actors at the international level (Karlsson-Vinkhuyzen & Vihma 2009, p. 414; Kahler 2000). A low degree of precision and bindingness can facilitate policy learning and persuasion. In contrast to hierarchical, command-and-control type regulations, soft regulatory practices usually involve a large number of state and non-state actors in policymaking. The more the actors affected by a policy have a say in decisionmaking, the more likely they are to accept the policy outcome to be implemented, even if their interests may not have been fully accommodated. Therefore, the inclusion of interested stakeholders typically enhances not only the quality of regulation but also its societal acceptance and its normative legitimacy. In effect, this can boost compliance behavior.

With soft regulation being on the rise, the EU has provided a rich laboratory for testing its effectiveness and legitimacy, particularly in the field of social and environmental policy. Since the 1970s, the EU has experimented with what has been referred to "new environmental policy instruments (NEPIs)" (cf. Jordan *et al.* 2003, p. 5). Before the

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Single European Act, environmental regulation was strongly linked to the imperatives of establishing a single European market. Harmonization of product standards and production processes through generally binding emission limits was an essential step toward the abolition of competitive (dis)advantages to industry stemming from environmental legislation. Toward the end of the 1980s, however, this command-and-control approach became increasingly challenged. Successive enlargements of the EU increased heterogeneity among member states with regard to their administrative capacities, regulatory traditions, environmental conditions, and economic constraints to effectively incorporate the *acquis communautaire* into their domestic regulatory regimes. Moreover, the persisting compliance problems of certain member states with EU environmental legislation generated skepticism regarding the effectiveness of harmonization, calling for the introduction of more flexible regulatory approaches that allow member states to adapt EU laws to their specific national or subnational conditions, or both (Börzel 2003). The EU responded to these regulatory challenges with the increasing use of soft, less coercive, market-based policy instruments and procedural directives that provide significant leverage to domestic regulators in the implementation process (cf. Homeyer *et al.* 2000; Scott 2000; Zylicz & Holzinger 2000).

However, in spite of strong normative assumptions regarding the alleged effectiveness of soft forms of regulation to facilitate better regulation in heterogeneous environmental regimes, the actual capacity of novel regulatory approaches is rarely being addressed in a systematic way. Evidence from studies on employment policies, social protection, public health, and taxation offer rather moderate appraisals regarding the capacity of soft policy instruments to achieve tangible goals and bring about policy change (Hervey 2008; Kröger 2008; Scott & Trubek 2002). This seems to be even more the case in countries with rather weak regulatory capacities, such as the southern and eastern EU members (O'Hagan 2004; Börzel 2009) or non-OECD (Organisation for Economic Co-operation and Development) states (McAllister *et al.* 2010; Van Rooij 2010). Recent studies cast serious doubts on the effectiveness of soft policy instruments, particularly with regard to fostering member state regulatory capacity, pointing to the need for hard regulation to make soft regulation work (Héritier & Lehmkuhl 2008; Börzel 2009; Héritier & Rhodes forthcoming).

In this article we seek to assess the effectiveness of soft regulation in EU environmental policymaking by analyzing the experience of four member states – Hungary, Poland, Romania, and Greece – in the application of the Directive on Integrated Pollution Prevention and Control (IPPCD). Regulation of this field is of utmost importance for virtually all producing sectors in the EU, which comprises more than 50,000 industrial installations in the 27 member states (ENTEC 2007). This policy area is characterized by a gradual departure from traditional command-and-control instruments based on the setting of uniform, legally binding emission limit values (ELVs) in a wide range of sectoral industrial activities to less coercive instruments. The IPPCD, enacted by the EU in 1996, moved away substantially from media-specific regulatory approaches to pollution abatement.[1] It introduced an integrated approach that incorporates a single permit system covering all pollutant activities of industry in water, air, and land. Instead of legally prescribing uniform ELVs, the IPPCD provides for indicative non-binding emissions limits to be defined in co-regulatory processes on the basis of best available technologies (BAT), taking into account local environmental, technical, and economic circumstances. These components of the Directive provide an excellent

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basis for the assessment of the effectiveness of softer regulation in the area of environmental policies in the EU.

The article proceeds in five steps. In the next section we provide a brief overview of the main procedural and substantial characteristics of the IPPCD in order to identify the core institutional conditions and requirements for the effective implementation of and compliance with the Directive. We argue that the application of non-binding requirements of the IPPCD requires significant cognitive, material, and political capacities of both state regulators and industry in order to internalize those requirements into their domestic regulatory traditions and practices. In Section 3 we explore the link between compliance with soft regulatory instruments and capacity of both state regulators and affected industry as the most pertinent explanatory factor that accounts for the effective application of the IPPCD at the domestic level. In Section 4 we analyze the experience from the application of the IPPCD in four member states of the EU. Contrary to conventional expectations, our analysis reveals that instead of facilitating a smooth adoption and adaptation to EU regulatory requirements, soft regulation might lead to widespread non-compliance and the "hollowing out" of the permitting procedure. In the final section we revisit our theoretical argument in the light of recent proposals of the European Commission for a thorough reform of the IPPCD with the aim of reducing discretion of local regulators and strengthening the binding effects of BAT-based ELVs.

2. Soft regulation in EU pollution prevention and control

Being largely modeled on the UK's Pollution Prevention and Control (PPC) regulatory regime (Jordan 2004), the IPPCD is pivotal in the EU's turn toward the systematic use of soft environmental regulation. The Directive was enacted in a period of considerable skepticism regarding the capacity of new member states from Central Eastern Europe (CEE) to fully adopt the EU's rather demanding and costly regulatory requirements. Experience from previous enlargements with the inclusion of member states from Southern Europe (SE), with limited economic potential and weak administrative resources to apply the environmental *acquis*, exacerbated the quest for flexibility in the application of environmental standards. Soft regulatory instruments appeared as a promising alternative to hard harmonization requirements given the increasing economic, social, and ecological heterogeneity of the EU (Holzinger 1999).

Instead of defining uniformly binding ELVs, the IPPCD provides for indicative non-binding emission limits based on BAT as a reference basis for domestic regulators issuing permits to industry. Single permits to industry are expected to be based on BAT defined at sectoral and subsectoral levels. Definition of BAT is delegated to ad hoc regulatory sectoral and subsectoral Technical Working Groups (TWGs). TWGs comprise a large number of actors, including representatives from the Commission, member state governments, industry associations, individual firms, environmental organizations, research institutes, universities, and national and EU environmental agencies (Koutalakis 2008). These networks facilitate the dissemination of information on BAT based on the benchmarking of best practices described in regularly updated BAT Reference Documents (BREFs). BREFs are not legally binding. They are considered to be soft policy instruments because they provide an indicative basis for the adoption of ELVs to be incorporated into single permits issued by domestic regulators to each industrial installation. This

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decentralized system of permissions allows considerable leverage for domestic authorities to interpret BAT and define ELVs according to local environmental and geographical conditions and the precise technical features of the installation. The technical nature of BREFs and the lack of expertise on the part of the Commission to evaluate the compatibility of BAT-based ELVs with broad environmental objectives illuminate the considerable policy relevance of the TWGs as regulatory networks (Bohne 2008b; Lange 2008).

Although there is considerable room for flexibility in the application of the IPPCD requirements, there are substantial barriers for domestic regulators to activate them in a manner that is beneficial for domestic industry and at the same time guarantees environmental protection. The lack of precision and clarity of BAT requirements requires considerable cognitive and administrative capacities of domestic regulators, such as information regarding the emission performance of individual plants, their technologies, and the absorption capacity of local ecosystems. The accuracy of these data often depends on the cooperation of permit authorities with individual industrial operators and representatives of sectoral organizations that gather data and expertise regarding the performance of individual plants and sectoral BATs. The same holds for the application of the single permit that requires merging numerous environmental permits for industrial emissions on air, water, and soil into one and the coordination of a large number of administrative units from different levels of government responsible for specific permit parameters.

3. Applying soft regulatory instruments: The role of domestic actors' capacities

Previous research identifies a number of crucial institutional conditions that facilitate efficient and effective application of EU environmental law at the domestic level. The capacity of the state to adopt, monitor, and enforce legally binding regulation features prominently in the implementation and compliance literature (Haas 1964; Chayes & Chayes 1993; Jänicke *et al.* 1997). The literature identifies two fundamental properties of state capacity. The first focuses on *intra-state* relations; that is, the internal coherence within the state that can enable the administrative apparatus to formulate and implement sound policies (Börzel *et al.* 2011). Administrative capacity refers to the amount of material and cognitive resources available to state actors to effectively monitor and enforce regulation (Perkins & Neumayer 2007). The second aspect concerns *state–society* relations; that is, linkages between non-state and state actors (Migdal 1988). Political capacity of the state refers to the ability of state actors to institutionalize cooperative arrangements with societal actors in order to pool resources and share the costs of compliance (Kohler-Koch 2000). Both types of state capacity crucially influence the preferences of domestic actors in favor of effective implementation. The capacity of the state to commit essential material or cognitive resources is a crucial determinant of successful application of EU law at the domestic level. Those resources allow state regulators to take autonomous action and cope with political and economic costs inherent in law enforcement to third parties.

State capacity may provide positive incentives to non-state actors to cooperate under the threat of unilateral governmental intervention in regulatory matters (Héritier & Lehmkuhl 2008). However, non-state actors' capacities also condition both the

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willingness and the capacity of the state to effectively apply costly policies. Environmental organizations act as effective watchdogs through activism, awareness actions, and litigation. The same holds for business. Through peer review of compliance performance between competitors individual firms and associations enhance monitoring and enforcement capacities of domestic regulators (Börzel 2003; Koutalakis 2004). Most notably, business actors provide essential cognitive resources that strengthen the regulatory capacity of the state. Technically complex issues characterized by uncertainty often require frequently changing cognitive and material resources for effective regulation, which state actors often do not have and lie with industry as the primary rule target. This is particularly the case in highly technical areas where the state depends on individual producers for crucial regulatory information related to product characteristics and production processes. As a result, the capacity of state to act autonomously is considerably impaired and depends on private actors' incentives and capacity to commit their resources to regulatory policymaking.

The literature offers an abundance of factors that affect the willingness and ability of private actors to engage in public policymaking. Historical institutionalists emphasize the role of domestic pre-existing policy traditions of interest intermediation and state–business relations as significant determinants of the political capacity of the state and private interests to engage in cooperative policymaking in order to pool resources and share economic and administrative costs of adjustment (Kohler-Koch 2000). Others point at specific economic gains that shape private actors' preferences in favor of or against alignment of domestic regulatory standards with the EU. Competitive pressures, market integration, and specific sectoral characteristics condition private actors' incentives (Andonova 2003; Espach 2006; Thornton *et al.* 2009; Koutalakis 2010). Industrial actors who face high competitive pressures stemming from low productivity and low investment potential in BAT are more likely to exert pressure on their governments to negotiate transition periods in the application of costly regulatory requirements or simply not properly apply them at the phase of domestic monitoring and enforcement. In the absence of strong domestic and transnational associations that can form effective coalitions and reduce transaction costs inherent in collective action and generate trust between their members, business actors might face serious disincentives to engage with essential resources in the policymaking process. The role of transnational associations and coalitions between companies is therefore the second most important factor that enables the effective diffusion of regulatory practices and rules into candidate member states. In exchange for their resources, private actors may gain influence on regulatory policy outcomes, reduce implementation costs, increase their control over other actors' compliance behavior, and gain competitive advantages. These benefits, however, do not always outweigh the costs of getting involved in public policymaking. Lobbying and informal contacts may be less resource intensive and more beneficial for industry. Moreover, information asymmetries may create disincentives to industry to share regulatory information that would not be otherwise accessible to competitors and regulatory authorities. Finally, firms may face difficulties in justifying the commitment of resources to their investors unless they can demonstrate tangible effects on their overall corporate performance. In our empirical research we demonstrate that the above dimensions of state and non-state actors' capacities are crucial determinants for the effective application of soft regulatory instruments incorporated into the IPPCD.

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4. The application of the IPPCD in Hungary, Poland, Romania, and Greece

This article draws on empirical findings from an exploratory study of the implementation of the IPPCD in Hungary, Poland, Romania, and Greece.[2] The article is based on extensive interviews with stakeholders from three groups; namely, public regulators directly dealing with the application of the directive, industrial actors from individual firms and industrial associations regulated by the IPPCD, and independent policy experts in the field of industrial pollution. Our aim was to identify conditions that facilitate a smooth application of the IPPCD soft regulatory instruments at the domestic level. The emphasis was on the effective implementation (output) of the IPPCD rather than on compliance (rule-consistent behavior).[3]

The selection of countries was based on their alleged weak economic, social, and administrative conditions to apply costly EU policies and the assumptions made by parts of the literature that the application of soft law provides the necessary flexibility to accommodate these conditions. Instead of systematically comparing weak with strong member states of the EU, we focus on countries where soft regulation is mostly needed to facilitate their adjustment to EU environmental standards because they lack the necessary capacities. Hungary, Poland, Romania, and Greece feature favorable conditions for an exploratory assessment of the normative claims attached to soft regulatory instruments. Differing in their political, economic, and social structures, the four countries are equally characterized by limited economic and administrative capacities and weak industrial competitiveness, which significantly impair their chances to effectively apply demanding EU policies. Our empirical analysis demonstrates that contrary to conventional expectations, the effectiveness of soft regulatory instruments is considerably impaired in those countries in which they are most needed; namely, countries with a weak industrial base and economic and administrative shortcomings to apply costly environmental standards.

4.1. Supply side: Weak state capacity

The application of the IPPCD has been an immense challenge for domestic regulators in SE and CEE member states. In stark contrast to most highly industrialized Western European states that have established, from the 1980s, horizontal and vertical coordination structures for regulating industrial pollution, all countries under investigation are latecomers in this respect (Bohne 2006). In Hungary, Poland, and Romania, prior to the introduction of the IPPCD, horizontal coordination between different administrative units with media-specific regulatory capacities was virtually absent (Buzogany 2009a,b; Guttenbrunner 2009). In Greece, legalism and a legacy of ineffective administrative apparatus plagued with clientelism created an unfavorable institutional environment for the enforcement of soft regulatory approaches emanating from the EU (Koutalakis 2009). The legacy of weak enforcement capacities of public regulators generated controversy among major stakeholders and the public over private involvement in environmental policymaking. The latter is often perceived as the institutionalization of privileged access of powerful industrial groups to decisionmaking. The dominant perception that law enforcement is not negotiable reflects the wider suspicion among the public that soft regulation is just a refined way of circumventing compliance with legal obligations (Koutalakis 2009).

In all countries under investigation, the introduction of the IPPCD represents a profound case of policy misfit between precise policy requirements for the application of

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the single permit system and pre-existing domestic regulatory traditions. Prior to the transposition of the IPPCD, pollution prevention was based on media-specific legislation. The predominant approach relied on detailed descriptive regulations that did not provide any flexibility to domestic regulations to define precise ELVs according to local economic and environmental conditions (World Bank 2007, p. 31). The *pre*scriptive rather than *de*scriptive regulatory style of the IPPCD posed significant challenges to state regulators in interpreting its legal provisions. The flexible definition of BAT-based ELVs requires considerable cognitive capacities from regulators to interpret vague legal concepts in precise legal requirements attached to individual permits. Guidance documents such as BREFs were alien to the legal tradition and regulatory practice as they do not hold immediate direct legal effects but do dilute the role played by the state as the main regulator.[4] Few guidance documents have been drawn up to sustain implementation and very few of them have been available to regulated industry.[5]

At the initial stage of the application of the IPPCD, the most pressing problem was weak administrative capacities to undertake preparatory actions for the participation of national representatives in TWGs and the elaboration of national sectoral reports on BAT-based ELVs. Data on industrial installations covered by the Directive and the state of available technologies in several industrial sectors were not available. The same holds for precise data on the environmental performance and available technologies of individual plants. The lack of essential knowledge and expertise generated considerable problems in defining precise ELV parameters that correspond to different geographical and ecological conditions. This has led to the proliferation of troubles in issuing, monitoring, and enforcing the individual permits based on these parameters. Understaffing and weak inter-institutional standing of the coordination units within the Ministries of Environment hindered the timely onset of policy planning. In Greece, the Ministry of Environment was seriously understaffed, with only 12 full time employees (Koutalakis 2009). In Hungary, the IPPCD coordination was initially placed under the auspices of a non-executive Institute for Environmental Management without adequate reach in the high ranks of political decisionmaking (Buzogany 2009a). In Romania, neither the national nor the regional regulators consolidated a role in the application of the IPPCD until the country's EU accession. Even after accession, the latter remained weak in terms of staffing and expertise (European Commission 2005).[6]

Similar deficiencies emerged regarding coordination between different ministries involved in the process of issuing permits to industry. The application of integrated and cross-media-based requirements of the IPPCD in countries with weak structures and patterns of inter-ministerial cooperation generated controversy between different units that regulate water, air, and soil pollution over control of substantial and procedural aspects related to implementation. In Greece, antagonism between four different ministerial departments that were responsible for issuing six different environmental permits resulted in administrative inertia and blocked the timely transposition and the actual application of the Directive. In Poland and Hungary, weak coordination capacities resulted in significant problems between central and regional regulators that were responsible for issuing permits to industry (World Bank 2007).[7]

4.2. Demand side: Weak industrial actors

While state regulators faced considerable capacity shortcomings, the potential of non-state actors to compensate weak administrative capacities was also limited. Business

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actors had to shoulder the major part of the implementation costs involved in the application of BAT requirements and improve emission performance of their installations. Although there were some exceptions, most industrial sectors were severely affected by single permit requirements, suffering from outdated infrastructure and low investment potential. With some notable exceptions of highly competitive sectors that account for the bulk of foreign direct investment (FDI), such as pharmaceuticals and automobiles in Hungary, electronics in Poland, and cement in Greece, a survey carried out by the World Bank indicates that 43% of business respondents in our countries confirmed that the application of the IPPCD posed considerable implementation costs to their operations (World Bank 2007). In major sectors regulated by the IPPCD, including agriculture, the vast majority of plants are of small and medium size, with limited data regarding their own emission performance and low investment potential to resort to specialized consultants and environmental specialists (Mayer & Dragos 2005).

Lack of mutual trust between weak representative associations hindered the pooling of resources and sharing of implementation costs (Greenspan Bell 2004). Both state regulators and regulated industry showed only limited inclination to compensate for their weaknesses by engaging in mutual cooperation. Industry was skeptical toward any procedure directly or indirectly related to data provision for regulatory purposes (Justice and Environment 2007; Koutalakis 2009). Business actors have had little confidence in the capacities of the state to provide a level playing field because capacity shortcomings of regional regulators prevented the uniform application of the IPPCD across different territorial units (Mayer & Dragos 2005). Despite the promotion of stakeholder involvement in the TWGs, there is little evidence that the IPPCD has actually strengthened the involvement of business actors in regulation.

Comparisons between the types of actors and the rates of their participation in various TWGs reveal that all four countries have limited involvement in the process of defining technical parameters and precise requirements for the application of BAT-based ELVs at the EU level. Although Hungary, with only 96 installations, has considerably fewer industrial installations reporting to the European Pollutant Emission Register (EPER) than Poland, with 459, overall participation in various TWGs is remarkably identical between the two countries in absolute numbers. Both countries score the highest rates of participation among their CEE counterparts, with 16 participants in various TWGs each accounting for the 17.6% of participants from that region. However, turning to the functional composition of national representation in various TWGs, we identify considerable variations between the two countries, especially regarding business participation. Hungarian business participation is mainly channeled through large industrial sectoral associations that dominate the respective sectors. The most prominent examples are the Hungarian Steel Association, Cement, the Association of the Hungarian Aluminium Industry, the Association of Foundries, the Federation of Food Industries, and the Environmental Committee of the Hungarian Pharmaceutical Manufacturers Association. In contrast, Polish participation is dominated by individual companies, the vast majority of which belong to joint venture schemes with the participation of foreign investors and large energy producer companies such as the electricity plants of Krakow and Belchatow.[8] In Greece, participation in TWGs is dominated by large enterprises, such as the Public Power Corporation (DEI), Hellenic Petroleum, TITAN Cement Company SA, and Aluminium of Greece, and there are only a few sectoral associations of small and medium enterprises from sectors such as metal processing, canning, chemicals, meat

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production, and steel. Not all these actors were receptive to IPPCD single permit requirements. There was a certain degree of uncertainty related to the binding character of BAT requirements for existing installations and costs involved for the upgrade of their infrastructure. While all four countries saw considerable awareness-raising action (e.g. conferences and workshops), representation in the TWGs continues to be low and is dominated by public regulators. The lack of essential cognitive resources in terms of knowledge and expertise has limited the potential of domestic industry to benefit from existing opportunities to influence important decisions at the EU level regarding the adoption of indicative BAT requirements. As a result, most TWGs reflect the state of BAT of advanced Northern and Western European industry (Koutalakis 2008).

5. The IPPCD recast: Will the pendulum swing back to hard regulation?

The IPPCD case manifests the structural limits in the application of soft regulatory approaches in highly heterogeneous territorial and institutional units such as the EU. The lack of capacity from both state and non-state actors coupled with the low degree of institutionalization of cooperative modes of environmental governance hindered the application of the Directive. In Greece, the transposition of the IPPCD was considerably delayed. It took almost four years after the deadline for transposing the Directive (30 October 1999) and a judgment by the European Court of Justice for delayed transposition (C-64/01) to persuade the Greek government to transpose the Directive in 2002. However, implementing measures were adopted much later by three ministerial decrees issued in 2002 and 2003 that define the detailed requirements for issuing single permits to new and existing industrial installations and the distribution of competencies between central government departments and subnational authorities sharing responsibilities over industrial pollution. In Romania and Greece, where the BAT principle is often regarded as being at odds with the domestic administrative culture, the BREFs were simply validated through command-and-control style ministerial orders rather than leaving them as open guidance documents.[9] Poland and Romania successfully applied for very long phase-in (transition) periods; as a result, one-third of the IPPCD industrial plants were granted periods of grace of up to 12 years (DANCEE 2003; Buzogany 2009b).[10] In Hungary, regional authorities managed to cope with the rapidly growing demand for issuing IPPCD permits despite high capacity shortcomings, but at the price of diluting the requirements of a truly integrated cross-media BAT-based permit (Kohlheb & Pataki 2002; Jávor & Németh 2007). As a result, the introduction of single IPPCD permits did not replace media-specific ones. Instead of simplifying permit procedures, the introduction of the IPPCD increased administrative burdens to industry (Jávor & Németh 2007).[11]

Moreover, in order to cope with the financial and administrative challenges of the IPPCD, a number of alternative regulatory approaches emerged. A substantial number of the highly polluting installations used the possibility to opt out from the IPPCD and be regulated by the Large Combustion Plants Directive,[12] which provides for generally binding ELVs for only air pollution.[13] All four countries also made extensive use of General Binding Rules (GBRs); that is, sectoral agreements that provide direct conditions or minimum standards with binding effects on state regulators and individual operators. This provides regulators with legal certainly regarding the precise ELVs applied to individual installations instead of specifying local environmental quality standards based

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on BAT (VITO *et al.* 2007). In practice, "old" regulatory tools such as GBRs provide the opportunity to circumvent BAT requirements because they facilitate the application of generally binding ELVs. The latter are preferable to local regulators because they are more transparent, consistent, and comparable.[14] Thus, countries with weak regulatory capacities include BAT merely as a formal reference to enact ELVs that are mostly identical to sectoral pollution standards prior to the introduction of the IPPCD (Justice and Environment 2007).[15]

In response to the implementation problems, the Commission initiated a series of infringement proceedings against member states, predominately from Southern and Eastern Europe.[16] Moreover, the Commission called for a "recast" of the IPPCD and a comprehensive reform and simplification of all the industrial emission policies. The outcome of this process was the adoption, in December 2007, of a proposal for a new Directive on Industrial Emissions.[17] The proposal conjoins seven existing directives on industrial emissions into a single "omnibus" directive. It underlines the importance of a better application of BAT. However, it proposes streamlining enforcement through the introduction of mandatory plant inspections and common enforcement procedures. In order to strengthen the role of BAT, it limits discretion of local regulators in specifying ELVs by requiring authorities to justify in detail those permits that do not reflect the BAT. Transparency shall be increased altogether as any deviation from BAT has to be made public. These measures will strengthen the role of the Commission because deviations from BAT are subject to its monitoring and final approval. Potentially, this might also lead to "hybridity" in governance arrangements (Trubek *et al.* 2005). In its first reading, the European Parliament introduced a proposal, the so-called European Safety Net, which provides for the Commission to establish minimum requirements for all industrial sectors based on BREFs. However, in the Council, the Commission's draft met considerable resistance from the UK as well as from SEs and CEE member states. The "dirty coal front" led by the UK, Spain, Italy, Poland, Greece, and the Czech Republic lobbied for more flexibility and the delay of stricter provisions for large combustion plants till 2020.[18] It remains to be seen whether the Commission will succeed in its attempts to harden the requirements of the IPPCD and whether this will be the beginning of a general turn away from soft regulation in the EU.

6. Conclusions

Although administrative and political capacities are crucial determinants of the effectiveness of monitoring and enforcing hard law, they are ever more necessary to make soft policy instruments work. Our empirical analysis demonstrates that contrary to conventional expectations, the effectiveness of soft regulatory instruments is considerably impaired in those countries in which they are most needed; namely, in countries with a weak industrial base and economic and administrative shortcomings to apply costly environmental standards. Despite their alleged potential for facilitating flexible adjustments to distinctive economic, administrative, and environmental conditions, weak member states face considerable limitations in the application of those instruments. In Greece, Poland, Hungary, and Romania, material and cognitive resources of domestic regulators as well as their capacity to mobilize the knowledge and expertise of non-state actors are crucial determinants of effective responses to single permit requirements of the IPPCD. Those countries share a regulatory tradition characterized by the preponderance

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of legalistic and hierarchical approaches to environmental policy. At the same time, their administrative and cognitive resources to adopt and enforce environmental legislation are limited. These findings resonate with other studies that also raise serious theoretical and empirical concerns regarding the applicability of soft regulation in heterogeneous regulatory regimes such as the EU (Lubbe-Wolff 2001; ENTEC 2007; Bohne 2008a,b).

Although soft regulation appears to be a panacea for regulatory ineffectiveness, its application is hindered by the very same institutional conditions that generate a growing demand for the departure from generally binding regulation. Non-hierarchical, private self-regulation or public–private co-regulation require a strong shadow of hierarchy to be effective (but see Börzel & Risse 2010). Widespread implementation gaps and lack of enforceability might further undermine the legitimacy of the EU, raising serious doubts about the future viability of current regulatory arrangements. In other words, hard and soft regulation reinforce rather than substitute for each other. In order to support the effective implementation of hard regulation, soft law requires state and non-state actor capacities, mutual trust among the stakeholders involved, and constant and credible external pressure by governments or the EU. These conditions are hardly met outside the OECD world of industrialized democracies. Future accession countries in the southeast of Europe have even weaker governance capacities than their CEE counterparts. This is also true for the EU's near abroad, where the EU has even less relied on non-state actors to draw states closer to the *acquis*. Thus, researchers and policymakers alike should mute their high-flying expectations set into soft law. Our findings underscore the importance of other means to increase the effectiveness of hard regulation, including capacity building oriented toward both state and non-state actors.

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Notes

1 Council Directive 96/61/EC of 24 September 1996 concerning integrated pollution prevention and control.

2 For detailed case studies see the country chapters in Börzel (2009).

3 We account for effectiveness using three different indicators: (i) the complete and correct incorporation of the IPPCD into national legislation and the amendment or repeal of prior conflicting national rules (formal implementation); (ii) the administrative infrastructure and resources provided to put the objectives of the policy into practice and to monitor the rule-consistent behavior of the target actors (practical application); and (iii) the competent

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authorities equipped to encourage or compel rule-consistent behavior of the target actors by effective monitoring, positive and negative sanctions, and compulsory corrective measures (monitoring and enforcement). Therefore, we assess neither the long-term impacts of the IPPCD on the environmental performance of regulated industry nor the potential policy-learning effects facilitated by soft regulatory instruments.

4 Interview, Public Official, General Inspectorate of Environment Protection and Water, Hungary, 3 December 2007.

5 Interview, Policy Expert at a Consulting Company, Warsaw, 9 October 2006; Interview, Public Official, Ministry of Environment and Water Management, Bucharest, 11 January 2007.

6 Interview, Public Official, Ministry of Environment and Water Management, Bucharest, 11 July 2007.

7 Interview, Public Official, General Inspectorate of Environment Protection and Water, Budapest, 3 December 2007.

8 Source: Own elaboration based on information provided by the European Integrated Pollution Prevention and Control (IPPC) Bureau (EIPPCB), available at http://eippcb.jrc.es/. See also Koutalakis (2008).

9 Interview, Public Official, Ministry of Environment and Water Management, Bucharest, 11 July 2007.

10 The Polish Steel Association successfully bargained for longer transition periods (until 2010) for receiving integrated permits and financed the training of public administrators to help strengthen the administrative capacities of state actors necessary to apply and enforce the IPPCD; Interview, Policy Expert with Industrial Association, Warsaw, 23 October 2005.

11 Interview, Policy Expert at a Consulting Company, Budapest, 29 November 2007.

12 Directive 2001/80/EC of the European Parliament and of the Council of 23 October 2001 on the limitation of emissions of certain pollutants into the air from large combustion plants.

13 Report on existing plants declared for eligibility under Article 4.4 of the Large Combustion Plants (LCP) Directive (http://dataservice.eea.europa.eu/dataservice/metadetails.asp?id=1086).

14 The option of using GBRs for implementation of the IPPCD is given in Article 9 (8).

15 European Environmental Bureau: IPPC Recast – Promising but Far From Perfect. Available from URL: www.eeb.org/publication/Metamorphosis-48-February2008.pdf, p. 10 (last accessed 12 Jun 2009).

16 European Commission: Commission Warns Nine Member States for Not Issuing Industrial Permits on Time, IP/08/704, Brussels, 6 May 2008. Available from URL: http://europa.eu/rapid/pressReleasesAction.do?reference=IP/08/704&format=HTML&aged=0&language=EN&guiLanguage=en (last accessed 20 Oct 2009). Environment: Commission Pursues Action Against 10 Member States over Industrial Permits. Available from URL: http://europa.eu/rapid/pressReleasesAction.do?reference=IP/09/175&format=HTML&aged=0&language=EN&guiLanguage=de (last accessed 20 Dec 2009).

17 Proposal for a Directive of the European Parliament and of the Council on Industrial Emissions (Integrated Pollution Prevention and Control) (Recast). Available from: http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=COM:2007:0844:FIN:EN:PDF (last accessed 20 Dec 2009).

18 European Voice: Pollution Proposals Face Opposition. Available from URL: http://www.europeanvoice.com/article/imported/pollution-proposals-face-opposition/64169.aspx (last accessed 5 Jul 2009).

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